Thompson's group F is nonamenable

Rewritten for readability from the OpenAI manuscript of the same title

September 23, 2026

Abstract

We prove that Thompson's group F is nonamenable. This confirms Geoghegan's conjecture and resolves the amenability problem for F.

1 The result and the shape of the proof

1.1 Statement

Thompson's group F consists of the increasing piecewise linear homeomorphisms of $[0,1]$ with finitely many linear pieces, breakpoints at dyadic rationals (numbers $k/2^n$), and slopes in $2^{\mathbb{Z}}$. The group operation is composition, written $hg = h \circ g$ (apply g first), and F is regarded as a discrete group.

A discrete group Γ is *amenable* if its bounded real-valued functions admit a positive normalized left-invariant mean: a linear functional $M : \ell^\infty(\Gamma;\mathbb{R}) \to \mathbb{R}$ with $M(1) = 1$, $M(\varphi) \geq 0$ whenever $\varphi \geq 0$, and $M(g \mapsto \varphi(hg)) = M(\varphi)$ for every $h \in \Gamma$. (The source writes G for a general group; we write Γ because G denotes a map in the appendix.) This viewpoint belongs to the classical theory of von Neumann and Day [19, 6].

Theorem 1.1. *Thompson's group F is not amenable.*

This is Geoghegan's 1979 conjecture, as recorded in [5, p. 227].

The only fact about amenability that is used. We use the forward implication of the Følner criterion [7]: *if Γ is amenable, then for every finite $S \subset \Gamma$ and every $\varepsilon > 0$ there is a nonempty finite $A \subset \Gamma$ such that*

$$\frac{|hA \triangle A|}{|A|} < \varepsilon \qquad (h \in S). \tag{1.1}$$

Here $hA = \{hg : g \in A\}$ and $\triangle$ is the symmetric difference. The modern left-translation formulation in [14, Theorem 1.2] bounds the sum over $h \in S$ of these ratios. Applying it with a smaller tolerance (say $\varepsilon/2$) gives each individual strict bound in (1.1). So, to prove Theorem 1.1, it suffices to find one finite $S \subset F$ and one constant $c_0 > 0$ such that $\max_{h\in S} |hA \triangle A|/|A| \geq c_0$ for *every* nonempty finite $A \subset F$. This is Proposition 4.2.

1.2 The one analytic input

Lemma 1.2 (Benyamini–Sternfeld)**.** *There are a real Hilbert space H, a Lipschitz map $f : B \to B$ on its closed unit ball B, and a constant $\delta > 0$ such that*

$$\|f(x) - x\| \geq \delta \qquad (x \in B).$$

Benyamini and Sternfeld prove this for every infinite-dimensional normed space [2, Theorem, part (3), p. 439]. Appendix A gives a self-contained construction on $L^2([0,1];\mathbb{R}^2)$ with $\delta = 1/2$. *Sanity check.* No such f exists when $\dim H < \infty$, since Brouwer's theorem gives a fixed point. This is why the proof below needs Hilbert-valued, not scalar, colours. Lemma 1.2 is used exactly once, in (4.6).

1.3 Proof at a glance

Key idea. Let f, L, δ be as in Lemma 1.2 (L a Lipschitz constant of f) and let D be a large integer. To each $g \in F$ we attach finitely many vectors $X_I(g) \in B$, indexed by intervals I. They satisfy an *exact nonlinear recursion* through f. Their pairwise inner products are also *exactly transported* by suitable elements of F. If a finite set $A \subset F$ were almost invariant under those elements, all these inner products would have nearly the same average over A. A two-line Hilbert-space computation would then turn the recursion into an approximate fixed point of f, which Lemma 1.2 forbids.

Objects (defined precisely in Sections 2 and 3).

- A *basic partition* is a finite partition of $[0,1]$ into dyadic intervals $[k2^{-r}, (k+1)2^{-r}]$. For $g \in F$ and large n, the image $gT^{(n)}$ of the uniform level-n partition is basic.
- Fix D *parent* intervals $I_1 < \cdots < I_D$ with gaps between them, inside $(0,1)$. Inside each parent I_i put the affine copy of the same family; this gives D^2 *descendants* $I_i \cdot I_j$.
- A *colouring* p assigns to every basic partition T a vector $p(T) \in B$ by recursion on the number of cells. If T is fine enough, $p(T) = f$(mean of the colours of the D pieces of T in $I_1, \dots, I_D$, each rescaled to $[0$. Otherwise $p(T) = 0$.
- $X_I(g)$ is the colour of the piece of $gT^{(n)}$ in I, rescaled to $[0,1]$.

Steps and dependencies.

(1) *Recursion identity (exact).* $X_{I_i}(g) = f(z_i(g))$, where $z_i(g) = \frac{1}{D}\sum_j X_{I_i \cdot I_j}(g)$. Uses Lemma 2.1(c) and the definition of p (Lemma 3.2).

(2) *Transport (exact).* If $h \in F$ maps I affinely onto I', then $X_I(g) = X_{I'}(hg)$ (Lemma 2.4). For every pair $I < J$ of our intervals with a gap between them, F contains an element $h_{I,J}$ mapping $I \to I_1$ and $J \to I_2$ affinely (Lemma 2.3). The finite set S of these elements is fixed before A.

(3) *Equal correlations.* Let $\eta = \max_{h \in S} |hA \triangle A|/|A|$ and $\alpha = \mathbb{E}_A\langle X_{I_1}, X_{I_2}\rangle$ (average over A). Then $|\mathbb{E}_A\langle X_I, X_J\rangle - \alpha| \le \eta$ for every such pair. Uses (2), the list of separated pairs (Lemma 3.3) and the counting Lemma 4.1.

(4) *Variance estimate.* With $m = \frac{1}{D}\sum_k X_{I_k}$, expanding $\|z_i - m\|^2$ and using (3) on all but a fraction $1/D$ of the inner products gives $\mathbb{E}_A\|z_i - m\|^2 \le 4/D + 4(1 - 1/D)\eta$; the unknown α cancels.

(5) *Contradiction with displacement.* By (1), $m = \frac{1}{D}\sum_i f(z_i)$, so $\delta^2 \le \|m - f(m)\|^2 \le \frac{L^2}{D}\sum_i \|z_i - m\|^2$. Averaging over A and inserting (4) gives $\delta^2 \le L^2(4/D + 4(1 - 1/D)\eta)$. For $D > 4L^2/\delta^2$ this is a positive lower bound on η independent of A (Proposition 4.2), contradicting (1.1).

The level n is chosen after A, so that all partitions involved are fine enough (Lemma 2.2). This is harmless because S and the final constant do not depend on n. Section 5 records two consequences that rely on separate companion theorems. Appendix A proves Lemma 1.2.

1.4 Context and prior work

This subsection is background and is not used in the proof.

Thompson introduced F in 1965; an early published construction appears in McKenzie and Thompson [11], with the identification explained by Cannon, Floyd, and Parry [5, pp. 215–216]. Two classical structural facts help explain the difficulty of the amenability problem. Brin and Squier show that F contains no nonabelian free subgroup [3, Theorem 3.1], so the familiar free-subgroup obstruction does not apply. Yet F is not elementary amenable [5, Theorem 4.10]: it lies outside the class generated by finite and abelian groups under subgroups, quotients, extensions, and directed unions. Neither fact decides ordinary amenability. Brown and Geoghegan construct a classifying space with finitely many cells in each dimension [4].

Moore proves tower lower bounds on the sizes of possible Følner sets [15, Theorem 1.1] and characterizes amenability of F by a scalar convex Ramsey property for finite rooted ordered binary trees [14, Theorem 3.1]. Guba improves densities of finite Cayley subgraphs [8] and derives restrictions on right-invariant means from a partition into seven diagram-defined classes [9]. Haagerup and Olesen prove that simplicity of the reduced group C^*-algebra of Thompson's group T would imply nonamenability of F [10, Theorem 5.5]. The proof below uses finite dyadic partitions and scalar correlations of Hilbert-valued colours.

There have also been claims of a complete resolution in both directions. Akhmedov's 2021 version claims nonamenability through a height-function criterion [1]. Shavgulidze claimed amenability [18]; Moore identified errors in that approach [12]. Moore separately withdrew an amenability claim after Akhmedov identified an error in Lemma 4.13 of that manuscript [13]. That withdrawal concerns a different work from his published Ramsey characterization cited above.

2 Dyadic partitions and affine transport

We use the standard dyadic-partition model of F; see [5, Section 2, Lemma 2.2]. Everything needed is proved here.

2.1 Definitions

- A *basic (dyadic) interval* is an interval $[k2^{-r}, (k+1)2^{-r}] \subseteq [0,1]$ with $r \in \mathbb{N} \cup \{0\}$ and $0 \le k < 2^r$.
- A *basic partition* is a finite family of basic intervals (*cells*) with union $[0,1]$ and pairwise disjoint interiors; cells may share endpoints. Its *mesh* is the greatest length of a cell. $T^{(n)}$ denotes the uniform partition into the 2^n cells of length 2^{-n}.
- A partition T *respects* an interval I if I is a union of cells of T.
- For a basic interval I, $s_I : [0,1] \to I$ is the increasing affine bijection, and $I \cdot J = s_I(J)$ for a basic interval J.
- If T respects I, its *normalized restriction* is $T_I = \{s_I^{-1}(C) : C \in T,\ C \subseteq I\}$.
- $I < J$ means $\max I < \min J$; this requires a positive gap. We then call I, J *strictly separated*. An interval is *internal* if both endpoints lie in $(0,1)$.
- $h \in F$ *carries I affinely onto I'* if $h|_I$ is affine and $h(I) = I'$. Since h is increasing, $h \circ s_I$ is then an increasing affine bijection $[0,1] \to I'$, so

$$h \circ s_I = s_{I'}. \tag{2.1}$$

- For $g \in F$, $gT^{(n)} = \{g(C) : C \in T^{(n)}\}$.

Example. For $I = [\frac{1}{4}, \frac{1}{2}]$ and $J = [\frac{1}{2}, 1]$ we get $I \cdot J = [\frac{3}{8}, \frac{1}{2}]$. The partition $T^{(3)}$ respects I, and $(T^{(3)})_I = T^{(1)}$.

2.2 Normalized restrictions

The next lemma is used to show that the colouring is well defined (Lemma 3.2) and to prove the recursion identity (4.5).

Lemma 2.1. *Let I, J be basic intervals and T a basic partition.*

(a) $I \cdot J$ is a basic interval contained in I, and $s_{I\cdot J} = s_I \circ s_J$.

(b) If T respects I, then T_I is a basic partition, with exactly as many cells as T has cells contained in I.

(c) If T respects both I and $I \cdot J$, then T_I respects J and

$$(T_I)_J = T_{I\cdot J}. \tag{2.2}$$

Proof. Write $I = [k2^{-r}, (k+1)2^{-r}]$, so $s_I(x) = 2^{-r}(x+k)$ and $s_I^{-1}(y) = 2^r y - k$.

(a) If $J = [k'2^{-r'}, (k'+1)2^{-r'}]$, then $s_I(J) = [(k2^{r'} + k')2^{-(r+r')}, (k2^{r'} + k' + 1)2^{-(r+r')}]$, a basic interval inside $s_I([0,1]) = I$. The map $s_I \circ s_J$ is increasing, affine, and maps $[0,1]$ onto $s_I(J) = I \cdot J$; such a map is unique, so it equals $s_{I\cdot J}$.

(b) Let $C = [j2^{-\ell}, (j+1)2^{-\ell}]$ be a cell of T with $C \subseteq I$. Then $\ell \geq r$, and $s_I^{-1}(C) = [(j - k2^{\ell-r})2^{-(\ell-r)}, (j - k2^{\ell-r} + 1)2^{-(\ell-r)}]$. Its left endpoint is an integer multiple of its length $2^{-(\ell-r)}$, and it lies in $s_I^{-1}(I) = [0,1]$, so it is basic. Because I is the union of the cells $C \subseteq I$, their images cover $[0,1]$. Their interiors are disjoint because s_I^{-1} is injective. Injectivity also gives the count of cells.

(c) By (a), $I \cdot J \subseteq I$, so the cells of T contained in $I \cdot J$ are among those contained in I. Since T respects $I \cdot J$, we get $J = s_I^{-1}(I \cdot J) = \bigcup\{s_I^{-1}(C) : C \in T,\ C \subseteq I \cdot J\}$, a union of cells of T_I. So T_I respects J. A cell $s_I^{-1}(C)$ of T_I lies in J exactly when $C \subseteq s_I(J) = I \cdot J$. Hence, using (a),

$$(T_I)_J = \{s_J^{-1}s_I^{-1}(C) : C \in T,\ C \subseteq I \cdot J\} = \{s_{I\cdot J}^{-1}(C) : C \in T,\ C \subseteq I \cdot J\} = T_{I\cdot J}. \qquad \square$$

2.3 Images of fine uniform partitions

The next lemma is used once, in the proof of Proposition 4.2, to choose the level n.

Lemma 2.2. *For each $g \in F$, the partitions $gT^{(n)}$ are basic for all sufficiently large n, and their meshes tend to zero. In particular, given a finite set $E \subset F$ and a finite family $\mathcal{J}$ of basic intervals, there is an n_0 such that $gT^{(n)}$ is basic and respects every member of $\mathcal{J}$ whenever $g \in E$ and $n \geq n_0$.*

Proof. **Step 1 (form of the pieces).** Let $0 = x_0 < x_1 < \cdots < x_N = 1$ be dyadic points containing all breakpoints of g, and let 2^{q_i} be the slope on $[x_{i-1}, x_i]$. Since $g(0) = 0$, we have $g(x_i) = \sum_{j \leq i} 2^{q_j}(x_j - x_{j-1})$, a dyadic rational. Hence on the i-th piece $g(x) = 2^{q_i}x + b_i$ with $b_i = g(x_{i-1}) - 2^{q_i}x_{i-1}$ dyadic.

Step 2 ($gT^{(n)}$ is basic for large n). Choose n so large that (i) every x_i is a multiple of 2^{-n}, (ii) $n \geq q_i$ for all i, and (iii) every b_i is an integer multiple of $2^{q_i - n}$. Each condition, once true, stays true for larger n, and there are finitely many pieces. By (i), every cell $[k2^{-n}, (k+1)2^{-n}]$ lies in one piece, say the i-th. Its image is $[k''2^{q_i-n}, (k''+1)2^{q_i-n}]$ with $k'' = k + b_i 2^{n-q_i}$, an integer by (iii). The length $2^{-(n-q_i)}$ is at most 1 by (ii), and the image lies in $[0,1]$, so it is a basic interval. As g is a homeomorphism of $[0,1]$, the images form a partition.

Step 3 (mesh). The mesh of $gT^{(n)}$ is $2^{\max_i q_i - n} \to 0$.

Step 4 (basic intervals with intersecting interiors are nested). Let I', I'' be basic of lengths $2^{-r'} \geq 2^{-r''}$. Every endpoint of $T^{(r')}$ is a multiple of $2^{-r''}$, so none lies in the interior of I''. Thus I'' lies in a single cell of $T^{(r')}$. If the interiors of I' and I'' meet, that cell must be I', so $I'' \subseteq I'$.

Step 5 (small mesh implies respect). Let T be a basic partition of mesh smaller than the length of a basic interval I. Take x in the interior of I and a cell $C \ni x$. Since C is a nondegenerate interval, its interior meets that of I. By Step 4 the two are nested, and C is shorter, so $C \subseteq I$. Hence the interior of I lies in the union of the cells contained in I. That union is closed and contained in I, so it equals I.

Step 6 (conclusion). For each $g \in E$, Steps 2, 3 and 5 give a threshold beyond which $gT^{(n)}$ is basic with mesh below the shortest member of $\mathcal{J}$, hence respects every member. Take n_0 to be the largest of these finitely many thresholds. □

2.4 Transport of pairs and exact covariance

The next lemma supplies the finite set S of Section 4.

Lemma 2.3. *If $I < J$ and $I' < J'$ are pairs of internal basic intervals, there is an $h \in F$ carrying I affinely onto I' and J affinely onto J'.*

Proof. **Step 1 (gaps).** The source pair leaves three gaps $[0, \min I]$, $[\max I, \min J]$, $[\max J, 1]$. Each has positive length because I, J are internal and $I < J$, and each has dyadic endpoints. The same holds for the target pair.

Step 2 (partition the gaps). Choose N with all endpoints of the six gaps multiples of 2^{-N}; then each gap is a union of cells of $T^{(N)}$, which are basic.

Step 3 (equalize). For each of the three positions, if the source gap has fewer cells than the corresponding target gap (or conversely), bisect one cell on the side with fewer cells. The two halves of a basic interval are basic and the number of cells increases by one. Repeat until the numbers agree.

Step 4 (define h). Now the list (cells of gap 1, I, cells of gap 2, J, cells of gap 3) and the corresponding target list are basic partitions of $[0, 1]$ with the same number of cells, with I, J and I', J' in matching positions. Let h map each source cell increasingly and affinely onto the corresponding target cell. Then h is an increasing homeomorphism of $[0, 1]$, piecewise linear with finitely many pieces, with breakpoints at dyadic cell endpoints and slopes that are ratios of two powers of two. So $h \in F$, and h is affine on I and on J with the required images. □

The next lemma is the exact covariance used in Step 1 of the proof of Proposition 4.2.

Lemma 2.4. *Let $g, h \in F$ and suppose h carries the basic interval I affinely onto the basic interval I'. If $gT^{(n)}$ and $(hg)T^{(n)}$ are basic and respect I and I', respectively, then*

$$\left((hg)T^{(n)}\right)_{I'} = (gT^{(n)})_I. \tag{2.3}$$

Proof. Since $hg = h \circ g$, the cells of $(hg)T^{(n)}$ are the sets $h(C)$ with C a cell of $gT^{(n)}$. As h is a bijection with $h(I) = I'$, we have $h(C) \subseteq I'$ if and only if $C \subseteq I$. For such C, (2.1) gives $s_{I'}^{-1} \circ h = s_I^{-1}$ on I, so $s_{I'}^{-1}(h(C)) = s_I^{-1}(C)$. The two families in (2.3) therefore coincide. □

3 The recursive colouring

3.1 Standing choices

Fix a real Hilbert space H, its closed unit ball B, and a Lipschitz map $f : B \to B$ as in Lemma 1.2. Fix $L > 0$ and $\delta > 0$ such that

$$\|f(x) - f(y)\| \le L\|x - y\|, \qquad \|f(x) - x\| \ge \delta \quad (x, y \in B). \tag{3.1}$$

Choose an integer $D \ge 2$ with

$$D > \frac{4L^2}{\delta^2}. \tag{3.2}$$

Choose internal basic intervals $I_1 < \cdots < I_D$, the *parents*. Such a family exists: take $2^r > 2D$ and $I_j = [(2j-1)2^{-r}, 2j\,2^{-r}]$. Then $\min I_1 = 2^{-r} > 0$, $\max I_D = 2D\,2^{-r} < 1$, and consecutive intervals are separated by a gap of length 2^{-r}.

3.2 Definition of the colouring

Definition 3.1. Define $p(T) \in B$ for every basic partition T by induction on the number of cells of T:

$$p(T) = \begin{cases} f\left(\dfrac{1}{D}\displaystyle\sum_{j=1}^{D} p(T_{I_j})\right), & \text{if } T \text{ respects every } I_j, \\ 0, & \text{otherwise.} \end{cases} \tag{3.3}$$

Lemma 3.2. *Definition 3.1 is a valid definition, and $p(T) \in B$ for all T.*

Proof. Suppose T respects every I_j. By Lemma 2.1(b), T_{I_j} is a basic partition whose number of cells equals the number of cells of T inside I_j. This is strictly less than the number of cells of T: since I_j is internal, the cell of T containing 0 is not contained in I_j. So $p(T_{I_j})$ is already defined and, inductively, lies in B. The mean of D points of the convex set B lies in B, the domain of f, and f takes values in B. In the other case $p(T) = 0 \in B$. □

Example. For the displayed parents $|I_j| = 2^{-r}$, the one-cell partition $T^{(0)}$ respects no I_j, so $p(T^{(0)}) = 0$. Next, $(T^{(r)})_{I_j} = T^{(0)}$ and $(T^{(2r)})_{I_j} = T^{(r)}$ for every j, so

$$p(T^{(r)}) = f(0), \qquad p(T^{(2r)}) = f(f(0)).$$

3.3 Parents, descendants, and the colour functions

Let

$$\mathcal{I} = \{I_i : 1 \le i \le D\} \cup \{I_i \cdot I_j : 1 \le i, j \le D\}.$$

The next lemma says which pairs of $\mathcal{I}$ can be transported by Lemma 2.3. It is used in Steps 1, 2 and 5 of the proof of Proposition 4.2.

Lemma 3.3. *Every member of $\mathcal{I}$ is an internal basic interval. The following pairs are strictly separated:*

(i) two parents I_k, I_l with $k \ne l$;

(ii) two siblings $I_i \cdot I_j$, $I_i \cdot I_l$ with $j \ne l$;

(iii) a descendant $I_i \cdot I_j$ and a parent I_k with $k \ne i$.

Proof. Parents are internal and basic by choice. $I_i \cdot I_j$ is basic by Lemma 2.1(a). Since $I_j \subset (0,1)$, we have $I_i \cdot I_j = s_{I_i}(I_j) \subset s_{I_i}((0,1))$, the interior of I_i, which lies in $(0,1)$. (i) holds by choice. (ii) holds because s_{I_i} is strictly increasing, so $\max I_j < \min I_l$ implies $\max(I_i \cdot I_j) < \min(I_i \cdot I_l)$. (iii) holds because $I_i \cdot I_j \subseteq I_i$, and I_i, I_k are strictly separated. □

Call $(n, g) \in \mathbb{N} \times F$ *admissible* if $gT^{(n)}$ is a basic partition respecting every member of $\mathcal{I}$. For every n, every $g \in F$, and every $I \in \mathcal{I}$, define

$$X_I^{(n)}(g) = \begin{cases} p((gT^{(n)})_I), & \text{if } (n,g) \text{ is admissible,} \\ 0, & \text{otherwise,} \end{cases} \qquad \varphi_{I,J}^{(n)}(g) = \langle X_I^{(n)}(g), X_J^{(n)}(g) \rangle. \tag{3.4}$$

So $X_I^{(n)} : F \to B$ and $\varphi_{I,J}^{(n)} : F \to [-1,1]$ are defined on all of F; the bound is Cauchy–Schwarz. By Lemma 2.2 (with $\mathcal{J} = \mathcal{I}$), for every finite $E \subset F$ there is a level n with (n, g) admissible for all $g \in E$.

4 A uniform boundary bound and the proof of the theorem

The finite set S. For every ordered pair $I < J$ of members of $\mathcal{I}$, use Lemma 2.3 (with target pair $I_1 < I_2$) to choose $h_{I,J} \in F$ carrying I affinely onto I_1 and J affinely onto I_2. Let S be the finite set of these elements. All choices so far (H, f, L, δ, D, the parents, p, S) are independent of the finite set A tested below.

The next lemma is used in Step 2 of the proof of Proposition 4.2.

Lemma 4.1. *For any nonempty finite $A \subset F$, any $h \in F$, and any $\varphi : F \to [-1,1]$,*

$$\left| \frac{1}{|A|} \sum_{g \in A} \varphi(hg) - \frac{1}{|A|} \sum_{g \in A} \varphi(g) \right| \le \frac{|hA \triangle A|}{|A|}. \tag{4.1}$$

Proof. Since $g \mapsto hg$ is a bijection $A \to hA$, $\sum_{g \in A} \varphi(hg) = \sum_{g' \in hA} \varphi(g')$. The terms with $g' \in hA \cap A$ cancel against the second sum, leaving $\sum_{hA \setminus A} \varphi - \sum_{A \setminus hA} \varphi$. Its absolute value is at most $|hA \setminus A| + |A \setminus hA| = |hA \triangle A|$. □

Proposition 4.2. *For every nonempty finite set $A \subset F$,*

$$\max_{h \in S} \frac{|hA \triangle A|}{|A|} \ge \frac{\delta^2/L^2 - 4/D}{4(1 - 1/D)} > 0. \tag{4.2}$$

Plan of the proof. Steps 1–2 show that all separated pairs have the same averaged correlation up to η. Steps 3–4 are pointwise identities and inequalities that involve f. Steps 5–6 bound $\mathbb{E}_A \|z_i - m\|^2$. Step 7 combines them.

Proof. Fix A and put

$$\eta = \max_{h \in S} \frac{|hA \triangle A|}{|A|}. \tag{4.3}$$

Let $SA = \bigcup_{h \in S} hA$. The set $A \cup SA$ is finite, so by Lemma 2.2 there is a single n such that (n, g) is admissible for all $g \in A \cup SA$. This n may depend on A. Fix it, omit the superscript (n), and write

$$\mathbb{E}_A \psi = \frac{1}{|A|} \sum_{g \in A} \psi(g), \qquad \alpha = \mathbb{E}_A \varphi_{I_1, I_2}.$$

Step 1 (transport identity). Let $I < J$ in $\mathcal{I}$, $h = h_{I,J}$, and $g \in A$. We claim

$$X_I(g) = X_{I_1}(hg), \qquad X_J(g) = X_{I_2}(hg).$$

Both (n, g) and (n, hg) are admissible, because $g \in A$ and $hg \in SA$. So $gT^{(n)}$ is basic and respects I, and $(hg)T^{(n)}$ is basic and respects I_1. Since h carries I affinely onto I_1, Lemma 2.4 gives $((hg)T^{(n)})_{I_1} = (gT^{(n)})_I$. Applying p and using the first case of (3.4) at both g and hg gives the first identity. The second is the same with (J, I_2). Note that h need not preserve the other intervals of $\mathcal{I}$; admissibility was required separately at g and at hg.

Step 2 (all separated correlations are nearly equal). We claim

$$|\mathbb{E}_A\langle X_I, X_J\rangle - \alpha| \le \eta \qquad \text{whenever } I, J \in \mathcal{I} \text{ are strictly separated.} \tag{4.4}$$

Assume first $I < J$ and let $h = h_{I,J}$. By Step 1, $\varphi_{I,J}(g) = \varphi_{I_1,I_2}(hg)$ for $g \in A$, so $\mathbb{E}_A\varphi_{I,J} = \frac{1}{|A|}\sum_{g\in A}\varphi_{I_1,I_2}(hg)$. Apply Lemma 4.1 to the function $\varphi_{I_1,I_2} : F \to [-1, 1]$ and then (4.3): $|\mathbb{E}_A\varphi_{I,J} - \alpha| \le |hA\triangle A|/|A| \le \eta$. If instead $J < I$, use the symmetry of the real inner product. Nothing is claimed for nested pairs.

Step 3 (recursion identity). For $g \in A$ put

$$m(g) = \frac{1}{D}\sum_{k=1}^{D} X_{I_k}(g), \qquad z_i(g) = \frac{1}{D}\sum_{j=1}^{D} X_{I_i\cdot I_j}(g) \quad (1 \le i \le D).$$

These are means of points of B, hence lie in B. We claim

$$X_{I_i}(g) = f(z_i(g)) \qquad (g \in A,\ 1 \le i \le D). \tag{4.5}$$

Let $T = gT^{(n)}$. By admissibility, T respects I_i and every $I_i \cdot I_j$. By Lemma 2.1(c), T_{I_i} respects every I_j and $(T_{I_i})_{I_j} = T_{I_i\cdot I_j}$. So the first case of (3.3) applies to T_{I_i}:

$$X_{I_i}(g) = p(T_{I_i}) = f\Big(\tfrac{1}{D}\sum_j p((T_{I_i})_{I_j})\Big) = f\Big(\tfrac{1}{D}\sum_j p(T_{I_i\cdot I_j})\Big) = f(z_i(g)).$$

Step 4 (pointwise displacement inequality). For every $g \in A$ (argument g suppressed),

$$\begin{aligned}
\delta^2 &\le \|m - f(m)\|^2 && \text{by (3.1), as } m \in B\\
&= \Big\|\frac{1}{D}\sum_{i=1}^{D}(f(z_i) - f(m))\Big\|^2 && \text{as } m = \tfrac{1}{D}\textstyle\sum_i X_{I_i} = \tfrac{1}{D}\sum_i f(z_i) \text{ by (4.5)}\\
&\le \frac{1}{D}\sum_{i=1}^{D}\|f(z_i) - f(m)\|^2 && \text{convexity of } \|\cdot\|^2\\
&\le \frac{L^2}{D}\sum_{i=1}^{D}\|z_i - m\|^2 && \text{by (3.1), as } z_i, m \in B.
\end{aligned} \tag{4.6}$$

The convexity step is $\|\frac{1}{D}\sum_i y_i\|^2 \le (\frac{1}{D}\sum_i\|y_i\|)^2 \le \frac{1}{D}\sum_i\|y_i\|^2$, by the triangle and Cauchy–Schwarz inequalities.

Step 5 (three averaged bounds). Fix i. We claim

$$\mathbb{E}_A\|m\|^2 \le \left(1 - \frac{1}{D}\right)(\alpha + \eta) + \frac{1}{D}, \tag{4.7}$$

$$\mathbb{E}_A\|z_i\|^2 \le \left(1 - \frac{1}{D}\right)(\alpha + \eta) + \frac{1}{D}, \tag{4.8}$$

$$\mathbb{E}_A\langle z_i, m\rangle \ge \left(1 - \frac{1}{D}\right)(\alpha - \eta) - \frac{1}{D}. \tag{4.9}$$

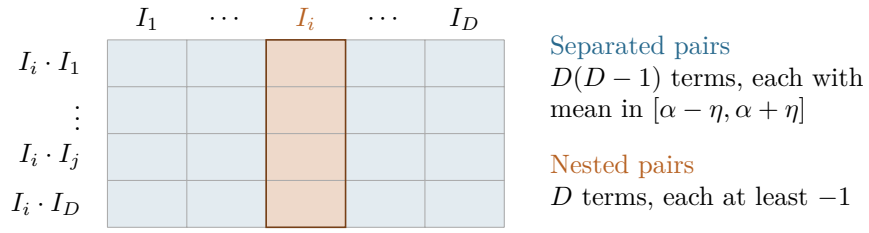


Figure 1: For fixed i, the (j,k) entry is $\langle X_{I_i \cdot I_j}, X_{I_k} \rangle$. Rows index the children of I_i and columns index all parents; the grid is schematic. Only column i contains nested pairs, a fraction $1/D$ of the D^2 terms.

Proof of (4.7). $\mathbb{E}_A \|m\|^2 = \frac{1}{D^2} \sum_{k,l} \mathbb{E}_A \langle X_{I_k}, X_{I_l} \rangle$. The $D(D-1)$ terms with $k \neq l$ involve strictly separated parents (Lemma 3.3(i)), so each is at most $\alpha + \eta$ by (4.4). The D terms with $k = l$ are at most 1 because the colours lie in B. Dividing by D^2 gives the bound.

Proof of (4.8). This is the same computation with the siblings $I_i \cdot I_j$, using Lemma 3.3(ii).

Proof of (4.9). $\mathbb{E}_A \langle z_i, m \rangle = \frac{1}{D^2} \sum_{j=1}^{D} \sum_{k=1}^{D} \mathbb{E}_A \langle X_{I_i \cdot I_j}, X_{I_k} \rangle$. The $D(D-1)$ terms with $k \neq i$ pair strictly separated intervals (Lemma 3.3(iii)), so each is at least $\alpha - \eta$. The D terms with $k = i$ pair a descendant with its own parent; each is at least -1 by the unit-ball bound. See Figure 1.

Step 6 (variance estimate). Expanding $\|z_i - m\|^2 = \|z_i\|^2 - 2\langle z_i, m \rangle + \|m\|^2$ and inserting (4.7)–(4.9),

$$\begin{aligned} \mathbb{E}_A \|z_i - m\|^2 &\leq 2\Big[\Big(1 - \frac{1}{D}\Big)(\alpha + \eta) + \frac{1}{D}\Big] - 2\Big[\Big(1 - \frac{1}{D}\Big)(\alpha - \eta) - \frac{1}{D}\Big] \\ &= \frac{4}{D} + 4\left(1 - \frac{1}{D}\right)\eta \qquad (1 \leq i \leq D). \end{aligned} \tag{4.10}$$

The terms in α cancel, so no assumption on the sign or size of α is needed.

Step 7 (conclusion). Average (4.6) over $g \in A$ and apply (4.10) to each i:

$$\delta^2 \leq \frac{L^2}{D} \sum_{i=1}^{D} \mathbb{E}_A \|z_i - m\|^2 \leq L^2 \left(\frac{4}{D} + 4\left(1 - \frac{1}{D}\right)\eta\right).$$

Hence $\delta^2/L^2 - 4/D \leq 4(1 - 1/D)\eta$. Dividing by $4(1 - 1/D) > 0$ (recall $D \geq 2$) gives (4.2). Its right-hand side is positive by (3.2) and does not depend on A. □

Proof of Theorem 1.1. Let $c_0 > 0$ be the right-hand side of (4.2). If F were amenable, (1.1) with the fixed finite set S and $\varepsilon = c_0$ would give a nonempty finite $A \subset F$ with $|hA \triangle A|/|A| < c_0$ for all $h \in S$. This contradicts Proposition 4.2. Hence F is not amenable. □

5 Consequences

Theorem 1.1 allows two companion results for nonamenable groups to be applied. Neither is used in the proof of nonamenability. Three symbols of the source are renamed here to avoid clashes: the Hilbert space is $\mathcal{K}$ (source: H), the similarity is V (source: S), and the generating set is Σ (source: S); the connection operator is written $\mathcal{C}_p$ (source: T_p). In Corollary 5.2 only, p denotes a probability, not the colouring of Section 3.

Corollary 5.1. *For every $\varepsilon > 0$, there are a separable complex Hilbert space $\mathcal{K}$ and a representation $\pi : F \to \mathrm{GL}(\mathcal{K})$ such that $\sup_{g \in F} \|\pi(g)\| \le 1 + \varepsilon$, but no bounded invertible operator V on $\mathcal{K}$ makes every $V\pi(g)V^{-1}$ unitary.*

Proof. The group F is countable, since each element is determined by finitely many dyadic breakpoints and slopes in $2^{\mathbb{Z}}$. It is nonamenable by Theorem 1.1. The conclusion follows from the companion unitarizability theorem [17, Theorem 1.1]. □

For a finite symmetric generating set $\Sigma \subset F \setminus \{\mathrm{id}\}$, let $\mathcal{G}_\Sigma$ be the simple undirected Cayley graph with edges $\{x, xs\}$, $s \in \Sigma$. In Bernoulli bond percolation on this graph, each edge is independently open with probability p. Write p_c for the threshold for existence of an infinite open cluster and p_u for the threshold for almost-sure uniqueness of the infinite open cluster. The connection kernel $\mathcal{C}_p(x, y) = \mathbb{P}_p(x \leftrightarrow y)$ acts on counting-measure $\ell^2(F)$, initially on finitely supported functions by summation over y; put $p_{2\to 2} = \sup\{p \in [0,1] : \mathcal{C}_p \text{ is bounded on } \ell^2(F)\}$.

Corollary 5.2. *For every such generating set Σ,*

$$\|\mathcal{C}_{p_c}\|_{2\to 2} < \infty, \qquad p_c < p_{2\to 2} \le p_u.$$

For each fixed $p \in (p_c, p_u)$, there are almost surely infinitely many infinite open clusters. Moreover, there are deterministic $p_c < p_1 < p_2 < 1$, depending on Σ, such that in the coupling with independent uniform $[0,1]$ edge labels U_e, opening e when $U_e \le p$, there are almost surely infinitely many infinite open clusters simultaneously for every $p \in [p_1, p_2]$.

Proof. Apply the companion percolation result [16, Corollary 1.2] to F, which is nonamenable by Theorem 1.1. □

Unverified step. The proofs of Corollaries 5.1 and 5.2 consist only of the two citations above. The source does not reproduce the cited statements and I could not consult the companion papers, so I cannot confirm that they say what is needed. What is needed is: for Corollary 5.1, that *every* countable nonamenable group has such a representation for each $\varepsilon > 0$; for Corollary 5.2, the displayed conclusions for *every* Cayley graph of a finitely generated nonamenable group. Both are strong general statements, so these two corollaries should be read as conditional on the companion papers.

A An explicit map with positive displacement

This appendix proves Lemma 1.2 directly, in the form of Proposition A.1 below ($\delta = 1/2$). It is the Hilbert-space instance of the Benyamini–Sternfeld theorem [2]. It is logically independent of Sections 2 to 4. *Notation.* Only H, B, f keep their meaning from the main text. All other letters $(a, b, q, c, d, r, t, \tau, \rho, G, \ldots)$ are defined afresh here. Compared with the source, the second curve parameter s is renamed τ, the operator A is renamed W, the constants K, M are renamed K_γ, M_γ, and $Jv(t)$ is written $v^\perp(t)$.

Proposition A.1. *Let $H = L^2([0,1]; \mathbb{R}^2)$, a real Hilbert space, and let $B = \{x \in H : \|x\| \le 1\}$. There is a globally Lipschitz map $f : B \to B$ such that*

$$\|f(x)\| = 1, \qquad \|f(x) - x\| \ge \frac{1}{2} \quad (x \in B).$$

Plan. We construct a Lipschitz map $G : B \to H$ with $\|G(x)\| \geq \rho/4 > 0$ everywhere and $G(x) = x$ when $\|x\| \geq 1/2$. Then $f(x) = -G(x)/\|G(x)\|$ works. The identity map fails only because it vanishes at 0. G modifies the identity inside a thin tube around a curve γ through 0, sliding the zero away along the curve. The curve's forward tail stays in a small ball yet never comes back close to itself (Lemma A.2). Lemma A.3 supplies rotations that keep the modification from being cancelled by the component transverse to the curve.

A.1 The curve

Define functions on $\mathbb{R}$ by

$$a(t) = \begin{cases} t/8, & t \leq 2, \\ 5/16 - (3-t)^2/16, & 2 < t < 3, \\ 5/16, & t \geq 3, \end{cases} \qquad \theta(t) = \begin{cases} 0, & t \leq 1, \\ (t-1)^2/2, & 1 < t < 2, \\ t - 3/2, & t \geq 2. \end{cases}$$

For $w \in [0, 1]$ put

$$v(t)(w) = (\cos(\theta(t)w), \sin(\theta(t)w)), \quad v^{\perp}(t)(w) = (-\sin(\theta(t)w), \cos(\theta(t)w)), \quad \gamma(t) = a(t)v(t).$$

Pointwise in w, $v(t)(w)$ and $v^{\perp}(t)(w)$ are orthonormal in $\mathbb{R}^2$; hence $\|v(t)\| = 1$ in H and $\|\gamma(t)\| = |a(t)|$. Also $\gamma(0) = 0$.

range	a	a'	θ	θ'
$t \leq 1$	$t/8$	$1/8$	0	0
$1 \leq t \leq 2$	$t/8$	$1/8$	$(t-1)^2/2$	$t-1$
$2 \leq t \leq 3$	$5/16 - (3-t)^2/16$	$(3-t)/8$	$t - 3/2$	1
$t \geq 3$	$5/16$	0	$t - 3/2$	1

The entries agree at $t = 1, 2, 3$ (for instance $a(2) = 1/4$ from both sides, $a'(2) = 1/8$, $a'(3) = 0$, $\theta(2) = 1/2$, $\theta'(1) = 0$, $\theta'(2) = 1$). We read off:

(F1) a, θ are C^1; $0 \leq a' \leq 1/8$ and $0 \leq \theta' \leq 1$; a' is $\frac{1}{8}$-Lipschitz and θ' is 1-Lipschitz.

(F2) $a \leq 5/16$ everywhere; a is strictly increasing on $(-\infty, 1]$; $a(t) \geq 1/8$ for $t \geq 1$; $a(t) \geq 1/4$ for $t \geq 2$; $a(t) = 5/16$ for $t \geq 3$.

(F3) $\theta = 0$ on $(-\infty, 1]$, θ is strictly increasing on $[1, \infty)$, and $\theta(t) = t - 3/2$ for $t \geq 2$.

(F4) The product $a\theta'$ vanishes on $(-\infty, 1]$, satisfies $0 \leq a\theta' \leq 5/16$, and is $\frac{7}{16}$-Lipschitz on $\mathbb{R}$. On $[1, \infty)$ we have $0 < a \leq 5/16$, $0 \leq \theta' \leq 1$, so $|a\theta'(t) - a\theta'(\tau)| \leq \frac{5}{16}|\theta'(t) - \theta'(\tau)| + |a(t) - a(\tau)| \leq (\frac{5}{16} + \frac{1}{8})|t - \tau|$. For $\tau \leq 1 \leq t$, compare with the value 0 at 1.

By (F3), $\gamma(t) = (t/8)\, v(0)$ for $t \leq 1$, a straight line. By (F2), $\|\gamma(t)\| \leq 5/16$ for $t \geq 1$.

The next lemma is used in the proof of Proposition A.1: it provides the tube U, the nearest-point parameter $t(x)$, and the Lipschitz control (A.1).

Lemma A.2. *The curve γ is injective, has a bounded Lipschitz derivative, and*

$$c := \inf_{t \in \mathbb{R}} \|\gamma'(t)\| = \frac{1}{8}.$$

Its unit tangent $u(t) = \gamma'(t)/\|\gamma'(t)\|$ is Lipschitz, $\langle u(t), v(t) \rangle \geq 0$, and $u(t) = v(t)$ for $t \leq 1$. There is a radius $0 < \rho < 1/32$ such that every point in

$$U = \{x \in H : \operatorname{dist}(x, \gamma(\mathbb{R})) < \rho\}$$

has a unique nearest point $\gamma(t(x))$. Writing $e(x) = x - \gamma(t(x))$, we have $e(x) \perp u(t(x))$, and

$$|t(x) - t(y)| \le 16\|x - y\| \quad \text{if } x, y \in U \text{ and } \|x - y\| < \rho. \tag{A.1}$$

Proof. **Step 1 (derivative in H).** For fixed $w \in [0,1]$, the $\mathbb{R}^2$-valued function $t \mapsto \gamma(t)(w)$ is C^1 by (F1), with derivative

$$g_w(t) = a'(t)\, v(t)(w) + a(t)\theta'(t)\, w\, v^{\perp}(t)(w).$$

We claim $|g_w(t) - g_w(\tau)| \le |t - \tau|$ for all $w \in [0,1]$. The unit vectors $v(t)(w)$ and $v^{\perp}(t)(w)$ have angle $\theta(t)w$, so each moves by at most $|\theta(t) - \theta(\tau)|w \le |t - \tau|$. By (F1), the first term changes by at most $|a'(t) - a'(\tau)| + \frac{1}{8}|t - \tau| \le \frac{1}{4}|t - \tau|$. By (F4), the second changes by at most $\frac{7}{16}|t - \tau| + \frac{5}{16}|t - \tau| = \frac{3}{4}|t - \tau|$. Hence, by the fundamental theorem of calculus, $|\gamma(t)(w) - \gamma(\tau)(w) - (t - \tau)g_w(\tau)| \le \frac{1}{2}|t - \tau|^2$ uniformly in w. Squaring and integrating over w gives the same bound in the norm of H. So γ is differentiable in H with

$$\gamma'(t) = a'(t)v(t) + a(t)\theta'(t)\, w\, v^{\perp}(t),$$

and $\|\gamma'(t) - \gamma'(\tau)\| \le |t - \tau|$. Write $K_\gamma = \mathrm{Lip}(\gamma') \le 1$.

Step 2 (speed). Since $v \perp v^{\perp}$ pointwise and $\int_0^1 w^2\, dw = \frac{1}{3}$,

$$\|\gamma'(t)\|^2 = a'(t)^2 + \frac{a(t)^2\theta'(t)^2}{3}.$$

For $t \le 2$ the first term is $1/64$. For $t \ge 2$ the second term is at least $(1/4)^2/3 = 1/48 > 1/64$ by (F2), (F3). For $t \le 1$ the second term vanishes. So $c = 1/8$, attained for $t \le 1$. Also $M_\gamma := \sup_t \|\gamma'(t)\| \le (1/64 + 25/768)^{1/2} < \infty$.

Step 3 (u and v). For $\|y\|, \|y'\| \ge \mu > 0$ one has

$$\Big\| \frac{y}{\|y\|} - \frac{y'}{\|y'\|} \Big\| \le \frac{\|y - y'\|}{\|y\|} + \frac{\big|\|y'\| - \|y\|\big|}{\|y\|} \le \frac{2}{\mu}\|y - y'\|. \tag{A.2}$$

With $\mu = c$, u is Lipschitz; write $K_u = \mathrm{Lip}(u)$. By Step 1's angle bound, v is 1-Lipschitz. Since $\langle v^{\perp}(t)(w), v(t)(w)\rangle = 0$ pointwise, $\langle \gamma'(t), v(t)\rangle = a'(t)$, so $\langle u(t), v(t)\rangle = a'(t)/\|\gamma'(t)\| \ge 0$. For $t \le 1$, $\gamma'(t) = \frac{1}{8}v(t)$, so $u(t) = v(t)$.

Step 4 (injectivity). Let $\mathrm{sinc}(\omega) = \sin(\omega)/\omega$ for $\omega \ne 0$ and $\mathrm{sinc}(0) = 1$. Integrating $\cos((\theta(t) - \theta(\tau))w)$ over $w \in [0,1]$ gives

$$\langle v(\tau), v(t)\rangle = \mathrm{sinc}(\theta(t) - \theta(\tau)), \qquad \|\gamma(t) - \gamma(\tau)\|^2 = a(t)^2 + a(\tau)^2 - 2a(t)a(\tau)\,\mathrm{sinc}(\theta(t) - \theta(\tau)). \tag{A.3}$$

Let $\tau < t$. If $t \le 1$, then $\|\gamma(t) - \gamma(\tau)\| = (t - \tau)/8 \ne 0$. If $t > 1$, then $\theta(t) > \theta(\tau)$ by (F3). Since $|\mathrm{sinc}(\omega)| < 1$ for $\omega \ne 0$, the unit vectors $v(\tau), v(t)$ are linearly independent (equality case of Cauchy–Schwarz). So $a(\tau)v(\tau) = a(t)v(t)$ would force $a(t) = 0$, contradicting $a(t) \ge 1/8$.

Step 5 (uniform separation). We claim

$$\sigma(d) := \inf_{|t - \tau| \ge d} \|\gamma(t) - \gamma(\tau)\| > 0 \qquad (d > 0). \tag{A.4}$$

Otherwise there are $\tau_n < t_n$ with $t_n - \tau_n \ge d$ and $\|\gamma(t_n) - \gamma(\tau_n)\| \to 0$. Pass to a subsequence with $\tau_n \to \tau_*$, $t_n \to t_*$ in $[-\infty, +\infty]$; $\tau_* \le t_*$.

- *Both finite.* Then $t_* - \tau_* \ge d$ and, by continuity, $\gamma(t_*) = \gamma(\tau_*)$, contradicting Step 4.

- $t_* = -\infty$ (so $\tau_* = -\infty$). Eventually both parameters are ≤ 1 and the distance is $(t_n - \tau_n)/8 \ge d/8$.
- $\tau_* = -\infty < t_*$. Then $\|\gamma(\tau_n)\| = |\tau_n|/8 \to \infty$ while $\|\gamma(t_n)\| = |a(t_n)|$ stays bounded.
- τ_* *finite,* $t_* = +\infty$. Then $\theta(t_n) - \theta(\tau_n) \to \infty$, so the sinc term in (A.3) tends to 0. Since $a(t_n) = 5/16$ eventually, the squared distances tend to $a(\tau_*)^2 + (5/16)^2 > 0$.
- *Both* $+\infty$. Eventually both are ≥ 3, so by (F2), (F3) and (A.3)

$$\|\gamma(t_n) - \gamma(\tau_n)\|^2 = 2(5/16)^2(1 - \operatorname{sinc}(t_n - \tau_n)) \ge 2(5/16)^2(1 - \sup_{\omega \ge d} \operatorname{sinc}(\omega)) > 0.$$

The supremum is less than 1: sinc is continuous, $\operatorname{sinc}(\omega) < 1$ for $\omega > 0$, and $|\operatorname{sinc}(\omega)| \le 1/\omega$.

Every case contradicts the choice of the sequences.

Step 6 (tangent estimate; choice of d and ρ). For a unit vector $\xi \in H$, the scalar function $s \mapsto \langle \gamma(s), \xi \rangle$ is C^1 with derivative $\langle \gamma'(s), \xi \rangle$, so $\langle \gamma(t) - \gamma(\tau) - (t-\tau)\gamma'(\tau), \xi \rangle = \int_\tau^t \langle \gamma'(s) - \gamma'(\tau), \xi \rangle \, ds$. Taking the supremum over ξ,

$$\|\gamma(t) - \gamma(\tau) - (t-\tau)\gamma'(\tau)\| \le \frac{K_\gamma}{2}|t-\tau|^2.$$

Choose $d > 0$ with $K_\gamma d \le c/2$. Pairing with $u(\tau)$, and using $\langle \gamma'(\tau), u(\tau) \rangle = \|\gamma'(\tau)\| \ge c$, we get for $|t - \tau| \le d$

$$|\langle \gamma(t) - \gamma(\tau), u(\tau) \rangle| \ge c|t-\tau| - \frac{K_\gamma d}{2}|t-\tau| \ge \frac{3c}{4}|t-\tau|. \tag{A.5}$$

Now choose $\rho > 0$ with

$$\rho < \frac{1}{32}, \qquad 4\rho < \sigma(d), \qquad \rho K_u < \frac{c}{4}. \tag{A.6}$$

Step 7 (nearest points exist and the residual is orthogonal). Let $x \in U$ and let $\gamma(t_n)$ be a minimizing sequence for $\operatorname{dist}(x, \gamma(\mathbb{R})) < \rho$. Late terms satisfy $\|x - \gamma(t_n)\| < 2\rho$, so they are pairwise less than $4\rho < \sigma(d)$ apart. By (A.4) their parameters differ by less than d. So the late t_n lie in a compact interval, and a convergent subsequence gives a parameter attaining the minimum. At any minimizing parameter t, $0 = \frac{d}{dt}\|x - \gamma(t)\|^2 = -2\langle x - \gamma(t), \gamma'(t) \rangle$, so $x - \gamma(t) \perp u(t)$.

Step 8 (Lipschitz parameter and uniqueness). Let $x, y \in U$ with $\|x - y\| < \rho$. Take any minimizing parameters t for x and τ for y, and write $x = \gamma(t) + e_t$, $y = \gamma(\tau) + e_\tau$, with $\|e_t\|, \|e_\tau\| < \rho$. Then $\|\gamma(t) - \gamma(\tau)\| < 3\rho < \sigma(d)$, so $|t - \tau| < d$. By Step 7, $\langle e_\tau, u(\tau) \rangle = 0$ and $\langle e_t, u(\tau) \rangle = \langle e_t, u(\tau) - u(t) \rangle$. Hence, by (A.5) and (A.6),

$$\begin{aligned} \|x - y\| \ge |\langle x - y, u(\tau) \rangle| &\ge |\langle \gamma(t) - \gamma(\tau), u(\tau) \rangle| - |\langle e_t, u(\tau) - u(t) \rangle| \\ &\ge \left(\frac{3c}{4} - \rho K_u \right) |t - \tau| \ge \frac{c}{2}|t - \tau| = \frac{|t-\tau|}{16}. \end{aligned}$$

Taking $y = x$ shows that the minimizing parameter is unique; call it $t(x)$. The same inequality is (A.1). □

A.2 Rotations

The next lemma is used in the lower bound (A.8). A residual $e \perp u(t)$ is carried to $R_t e \perp v(t)$, so that changing the coefficient along $v(t)$ cannot be cancelled by the residual.

Lemma A.3. *There is a family of orthogonal operators $R_t : H \to H$, Lipschitz in operator norm, such that $R_t u(t) = v(t)$ and R_t is the identity for $t \le 1$.*

Proof. Suppress t and put $\kappa = \langle u, v\rangle \geq 0$ (Lemma A.2). Define

$$W\xi = v\langle u, \xi\rangle - u\langle v, \xi\rangle \quad (\xi \in H), \qquad R = \mathrm{Id} + W + \frac{W^2}{1+\kappa}.$$

Step 1. W is skew-adjoint: $\langle W\xi, \zeta\rangle = \langle v, \zeta\rangle\langle u, \xi\rangle - \langle u, \zeta\rangle\langle v, \xi\rangle = -\langle \xi, W\zeta\rangle$.

Step 2 (u, v independent). $Wu = v - \kappa u$ and $Wv = \kappa v - u$, so $W^2u = (\kappa v - u) - \kappa(v - \kappa u) = -(1-\kappa^2)u$ and $W^2v = \kappa(\kappa v - u) - (v - \kappa u) = -(1-\kappa^2)v$. Thus on $\mathrm{span}\{u, v\}$, $W^2 = -(1-\kappa^2)\mathrm{Id}$ and $R = \mathrm{Id} + W - (1-\kappa)\mathrm{Id} = \kappa\,\mathrm{Id} + W$. This restriction is orthogonal, since by Step 1 $(\kappa\,\mathrm{Id} + W)^*(\kappa\,\mathrm{Id} + W) = \kappa^2\mathrm{Id} - W^2 = \mathrm{Id}$. Also $Ru = \kappa u + v - \kappa u = v$. On the orthogonal complement of the span, $W = 0$ and $R = \mathrm{Id}$. So R is orthogonal on H.

Step 3 (u, v dependent). Unit vectors with $\kappa \geq 0$ that are dependent are equal. Then $W = 0$, $R = \mathrm{Id}$, and $Ru = v$. By Lemma A.2 this is the case for $t \leq 1$.

Step 4 (Lipschitz). u, v are Lipschitz unit vectors, so $t \mapsto W_t$ and $t \mapsto \kappa(t)$ are Lipschitz and $\|W_t\| \leq 2$. Since $1 + \kappa \geq 1$, $t \mapsto 1/(1+\kappa(t))$ is Lipschitz and bounded. The formula for R then gives the claim, including where $u = v$. □

A.3 Proof of Proposition A.1

Fix ρ, U, $t(\cdot)$, $e(\cdot)$ from Lemma A.2 and R_t from Lemma A.3. Set

$$b(t) = \min\{a(t), -1/8\}, \qquad q(t) = b(t) - a(t) = -\max\{a(t) + 1/8, 0\}.$$

By (F2), $|q(t)| \leq 5/16 + 1/8 = 7/16$. Also $\mathrm{Lip}(q) \leq \mathrm{Lip}(a) \leq 1/8$. Let $\beta, \chi : [0, \infty) \to [0, 1]$ be the continuous piecewise linear functions

$$\beta(r) = \begin{cases} 1, & r \leq \rho/4, \\ 2 - 4r/\rho, & \rho/4 < r < \rho/2, \\ 0, & r \geq \rho/2, \end{cases} \qquad \chi(r) = \begin{cases} 1, & r \leq \rho/2, \\ 2 - 2r/\rho, & \rho/2 < r < \rho, \\ 0, & r \geq \rho. \end{cases}$$

Define $G : B \to H$ by $G(x) = x$ for $x \notin U$. For $x \in B \cap U$, write $t = t(x)$, $e = e(x)$, $r = \|e\| = \mathrm{dist}(x, \gamma(\mathbb{R}))$, and set

$$\begin{aligned} G(x) &= \gamma(t) + \beta(r)q(t)v(t) + \big((1 - \chi(r))\mathrm{Id} + \chi(r)R_t\big)e \\ &= x + \beta(r)q(t)v(t) + \chi(r)(R_t - \mathrm{Id})e. \end{aligned} \tag{A.7}$$

The two lines agree because $x = \gamma(t) + e$.

Proof of Proposition A.1. **Step 1 (G is globally Lipschitz on B).** We treat pairs $x, y \in B$ by cases.

(a) $x, y \in U$, $\|x - y\| < \rho$. By (A.1), $|t(x) - t(y)| \leq 16\|x - y\|$. Hence $\|e(x) - e(y)\| \leq \|x - y\| + \|\gamma(t(x)) - \gamma(t(y))\| \leq (1 + 16M_\gamma)\|x - y\|$, with M_γ from Step 2 of Lemma A.2. The distance $r(x) = \mathrm{dist}(x, \gamma(\mathbb{R}))$ is 1-Lipschitz. So each factor in the second line of (A.7) is Lipschitz on such pairs, with constants independent of the pair, and is uniformly bounded: $\beta(r), \chi(r) \in [0, 1]$; $|q| \leq 7/16$; $\|v\| = 1$; $\|R_t - \mathrm{Id}\| \leq 2$; $\|e\| < \rho$. A product of bounded Lipschitz factors is Lipschitz, so there is one constant for all such pairs.

(b) $\|x - y\| \geq \rho$. For all $x \in B$, $\|G(x)\| \leq 1 + 7/16 + 2\rho < 3/2$ (using $\rho < 1/32$). So $\|G(x) - G(y)\| \leq 3 \leq (3/\rho)\|x - y\|$.

(c) $x \in U$, $y \notin U$, $\|x-y\| < \rho$. Both cutoffs satisfy $\beta(r) \le \chi(r) \le 2(\rho-r)/\rho$ for $r < \rho$: for $r \le \rho/2$ the right side is ≥ 1, and for $r > \rho/2$ it equals $\chi(r)$ while $\beta(r) = 0$. Hence, from (A.7),

$$\|G(x) - x\| \le \frac{2(\rho-r)}{\rho}\Big(\frac{7}{16} + 2\rho\Big) = \Big(\frac{7}{8\rho} + 4\Big)(\rho - r).$$

Since $\operatorname{dist}(\cdot, \gamma(\mathbb{R}))$ is 1-Lipschitz and $\operatorname{dist}(y, \gamma(\mathbb{R})) \ge \rho$, we get $\rho - r \le \|x-y\|$. With $G(y) = y$, this gives $\|G(x) - G(y)\| \le (\frac{7}{8\rho} + 5)\|x-y\|$.

(d) $x, y \notin U$. G is the identity.

Step 2 (lower bound). We claim

$$\|G(x)\| \ge \rho/4 \qquad (x \in B). \tag{A.8}$$

Outside U: since $0 = \gamma(0)$, $\|G(x)\| = \|x\| \ge \operatorname{dist}(x, \gamma(\mathbb{R})) \ge \rho$.

Inside U, $r \le \rho/2$: here $\chi(r) = 1$, so $G(x) = (a(t) + \beta(r)q(t))v(t) + R_t e$. Since $e \perp u(t)$, R_t is orthogonal and $R_t u(t) = v(t)$, we have $\langle R_t e, v(t)\rangle = \langle e, u(t)\rangle = 0$ and $\|R_t e\| = r$. Thus $\|G(x)\|^2 = (a + \beta q)^2 + r^2$. If $r \le \rho/4$, then $\beta = 1$ and $a + q = b \le -1/8$, so $\|G(x)\| \ge 1/8 > \rho/4$. If $\rho/4 \le r \le \rho/2$, then $\|G(x)\| \ge r \ge \rho/4$.

Inside U, $\rho/2 \le r < \rho$: here $\beta(r) = 0$, and the last term in the first line of (A.7) has norm at most $r < \rho$. If $t > 1$, then $\|\gamma(t)\| = a(t) \ge 1/8$ by (F2), so $\|G(x)\| \ge 1/8 - \rho > \rho/4$. If $t \le 1$, then $R_t = \mathrm{Id}$ and $u(t) = v(t)$, so $G(x) = a(t)v(t) + e$ with $e \perp v(t)$, and $\|G(x)\| \ge r \ge \rho/2$.

Step 3 ($G(x) = x$ whenever $\|x\| \ge 1/2$). Outside U this is the definition. Inside U,

$$|a(t)| = \|\gamma(t)\| \ge \|x\| - r > 1/2 - \rho > 5/16.$$

Since $a \le 5/16$ everywhere, this forces $a(t) < -5/16 < -1/8$, hence $t \le 1$. Thus $q(t) = 0$ and $R_t = \mathrm{Id}$, and the second line of (A.7) gives $G(x) = x$.

Step 4 (the map f). Define $f(x) = -G(x)/\|G(x)\|$. By (A.8) and (A.2) with $\mu = \rho/4$, normalization is $(8/\rho)$-Lipschitz on the range of G, so f is globally Lipschitz by Step 1, and $\|f(x)\| = 1$. If $\|x\| < 1/2$, then $\|f(x) - x\| \ge 1 - \|x\| > 1/2$. Otherwise $G(x) = x$ by Step 3, so $f(x) = -x/\|x\|$ and $\|f(x) - x\| = 1 + \|x\| \ge 3/2$. □

Proposition A.1 gives Lemma 1.2 with $H = L^2([0,1]; \mathbb{R}^2)$ and $\delta = 1/2$.

Verification checklist

The steps that carry the proof, in the order a sceptical reader should check them:

1. **Order of choices.** S is fixed before A; only the level n depends on A; the final constant involves only δ, L, D. See the paragraph "The finite set S" and the opening of the proof of Proposition 4.2.
2. **Exact transport $X_I(g) = X_{I_1}(h_{I,J}g)$.** It rests on Lemma 2.4 (note $hg = h \circ g$ and left translation in (1.1)) and on admissibility at both g and hg. See Step 1 of Proposition 4.2.
3. **Which pairs are separated.** Cross-scale pairs (descendant of I_i with parent I_k, $k \ne i$) are separated, and only D of the D^2 mixed pairs are nested. See Lemma 3.3 and (4.9).
4. **Recursion identity $X_{I_i} = f(z_i)$.** It needs $(T_{I_i})_{I_j} = T_{I_i \cdot I_j}$ (Lemma 2.1(c)) and well-foundedness of p (Lemma 3.2). See Step 3 of Proposition 4.2.
5. **The algebra.** (4.7)–(4.9) imply (4.10) with α cancelling; then (4.6) and the rearrangement in Step 7.

6. **Existence of the transports.** Lemma 2.3 needs internal intervals and positive gaps so that all three gaps are nondegenerate.
7. **The displacement map (only if Lemma 1.2 is not taken from [2]).** Check the uniform separation (A.4), especially the case where both parameters tend to $+\infty$, then the lower bound (A.8), and Step 3 of the proof of Proposition A.1.
8. **Scope.** Corollaries 5.1 and 5.2 depend on companion papers not checked here; see the boxed note in Section 5.

References

[1] Azer Akhmedov. Non-amenability of R. Thompson's group F. arXiv:1310.4395v3, 2021.

[2] Y. Benyamini and Y. Sternfeld. Spheres in infinite-dimensional normed spaces are Lipschitz contractible. *Proceedings of the American Mathematical Society*, 88(3):439–445, July 1983. doi:10.1090/S0002-9939-1983-0699410-7.

[3] Matthew G. Brin and Craig C. Squier. Groups of piecewise linear homeomorphisms of the real line. *Inventiones Mathematicae*, 79(3):485–498, 1985. doi:10.1007/BF01388519.

[4] Kenneth S. Brown and Ross Geoghegan. An infinite-dimensional torsion-free FP_∞ group. *Inventiones Mathematicae*, 77(2):367–381, 1984. doi:10.1007/BF01388451.

[5] J. W. Cannon, W. J. Floyd, and W. R. Parry. Introductory notes on Richard Thompson's groups. *L'Enseignement Mathématique (2)*, 42(3–4):215–256, 1996. Primary article scan.

[6] Mahlon M. Day. Amenable semigroups. *Illinois Journal of Mathematics*, 1(4):509–544, 1957. doi:10.1215/ijm/1255380675.

[7] Erling Følner. On groups with full Banach mean value. *Mathematica Scandinavica*, 3:243–254, 1955. doi:10.7146/math.scand.a-10442.

[8] Victor Guba. Cayley graphs of R. Thompson's group F: New estimates for the density. *Journal of Combinatorial Algebra*, 9(1/2):145–168, 2025. doi:10.4171/JCA/110.

[9] Victor Guba. On zero-measured subsets of Thompson's group F. *Journal of Algebra*, 692:106–122, April 2026. doi:10.1016/j.jalgebra.2025.12.006.

[10] Uffe Haagerup and Kristian Knudsen Olesen. Non-inner amenability of the Thompson groups T and V. *Journal of Functional Analysis*, 272(11):4838–4852, 2017. doi:10.1016/j.jfa.2017.02.003.

[11] Ralph McKenzie and Richard J. Thompson. An elementary construction of unsolvable word problems in group theory. In W. W. Boone, F. B. Cannonito, and R. C. Lyndon, editors, *Word Problems: Decision Problems and the Burnside Problem in Group Theory*, volume 71 of *Studies in Logic and the Foundations of Mathematics*, pages 457–478. North-Holland, Amsterdam, 1973. doi:10.1016/S0049-237X(08)71914-1.

[12] Justin Tatch Moore. A note on Shavgulidze's papers concerning the amenability problem for Thompson's group F. arXiv:1102.0747v2, 2011.

[13] Justin Tatch Moore. Nonassociative Ramsey theory and the amenability of Thompson's group. arXiv:1209.2063v4, 2012. Withdrawn.

[14] Justin Tatch Moore. Amenability and Ramsey theory. *Fundamenta Mathematicae*, 220(3):263–280, 2013. doi:10.4064/fm220-3-6.

[15] Justin Tatch Moore. Fast growth in the Følner function for Thompson's group F. *Groups, Geometry, and Dynamics*, 7(3):633–651, 2013. doi:10.4171/GGD/201.

[16] OpenAI. Nonuniqueness of percolation on nonamenable quasi-transitive graphs. OpenAI Math Release preprint OAI:Nonuniqueness-of-percolation-on-nonamenable-quasi-transitive-graphs-September-24-2026, 2026. Corollary 1.2.

[17] OpenAI. Unitarizability implies amenability for discrete groups. OpenAI Math Release preprint OAI:Unitarizability-Implies-Amenability-for-Countable-Groups-September-23-2026, 2026. Theorem 1.1.

[18] E. T. Shavgulidze. About amenability of subgroups of the group of diffeomorphisms of the interval. arXiv:0906.0107v1, 2009.

[19] John von Neumann. Zur allgemeinen Theorie des Masses. *Fundamenta Mathematicae*, 13:73–116, 1929. doi:10.4064/fm-13-1-73-116.